1 Graphic Machinery

Overview

Menu	Production facility management → Current information → Graphic machinery Information management → Production overview → Graphic machinery HR management → Access control → Security control center
Transaction code	mpark
Function authorization	mpark

Purpose

The graphic machinery provides a clear presentation of important workplace and production information and the relevant contexts in space and time. The user can create and display different presentations (so-called layouts) of all organizational units (e.g. halls, departments).

You can use the *Graphic machinery* as a quick overview tool for the production manager. But the application can also be used in the shop floor to provide a current overview of production statuses on large screens.

Integration

In the graphic machinery, you can visualize up-to-date status information of objects from various HYDRA products:

- BDE/MDE: Machines, groups, line groups, key performance indicators (KPI), logged in operations
- MW: Terminals
- WRM/EMG: Resources (no file-based resources, no DNC resources)
- MPL: Material buffers
- ZKS: Accesses, access groups

You can only add and show KPIs if you enable the extensions mpark2kpi.

You can only add and show the operations logged on if you enable the extensions mpark2roptemp.

Requirements

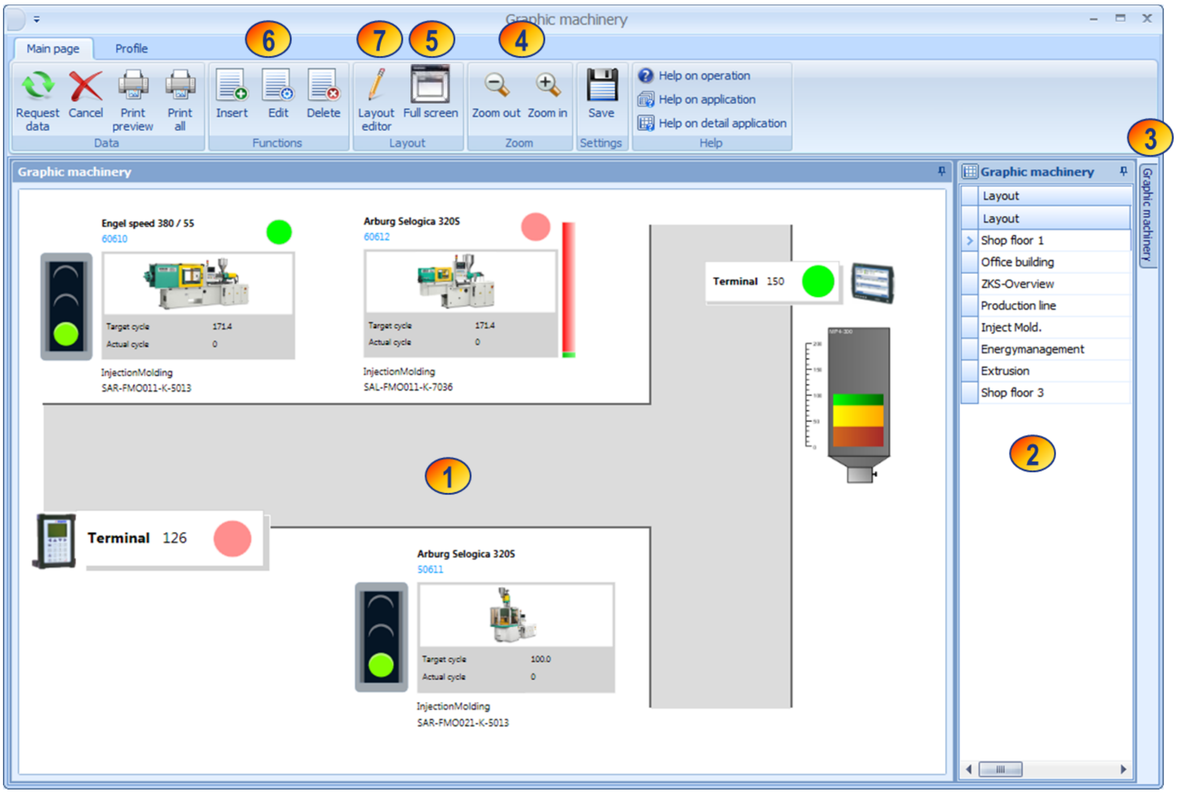
Set up the relevant objects you want to visualize on the Station Andon Board in the system master data.

Use the Layout editor to create the layouts required.

Structure of graphic machinery

You can use the *Graphic machinery* to show different layouts of different organizational units. You can generate these layouts according to your requirements.

To navigate through the layouts, use the table on the right hand side (2). The table contains the layout names and additional information on the different layouts. Left-click a table row to switch to the relevant layout which is then displayed in the layout panel (1).



(1) Layout panel

(2) List of available layouts

(3) Detail panel with information on the currently selected layout.

(4) Zooms in and out displayed layouts

(5) Starts the full screen mode; use the "Esc" button to cancel this mode.

(6) Editing functions to create a new layout and edit or delete an existing layout.

(7) Calls the the layout editor for graphic editing of the layout.

The application updates the statuses of machines, groups, terminals, etc. at regular intervals (every 30 seconds).

You can only request the *Help* function via F1 if the list of created layouts has the focus. In other cases, call up the *Help* function using the *Help on application* button.

Operation

Use the table on the right hand side of the screen (2) to navigate through the graphic machinery. Click a table row to switch to the requested layout.

Click the *Save settings* button to save the width of the table on the right hand side user-specifically. You cannot hide the table completely. Tip: You can "fold away" the table to the side using the respective button. You can find additional information on general operation in the documentation "Operation of the MES Operation Center".

If you right-click an icon in the layout panel (1), a context menu opens where you can switch to other applications, provided that you have the required authorization.

When you mouse over a template in a specific layout, a tooltip specific to each object opens showing additional information. You cannot configure the display time.

Details on the information displayed in a tooltip are described in the document MOC_GraphicMachinery2Edit.pdf, section *Notes on templates*.

Field descriptions

The following field descriptions refer to the table "Layout" and the detail view "Layout":

Layout

Layout name, e.g. "Hall 1"

Comment

Comment that describes the layout in more detail.

Sequence

If you specify a sequence/order and sort the table by the "order" column, the application always shows a particular layout by default when you call up the function. Please note that the defined order is not user-related.

Responsibility area

You can optionally assign a responsibility area to a layout. In this case, only users who are authorized for the entered responsibility area can use this layout.

Internal ID

The system assigns this number automatically every time you add a new layout. You cannot change this number.

File name

The layout configuration is saved in a file on the server. The system automatically assigns the file name when you add a new layout. You cannot change this file name.

Type

Always "N"

Modified by / Modified on

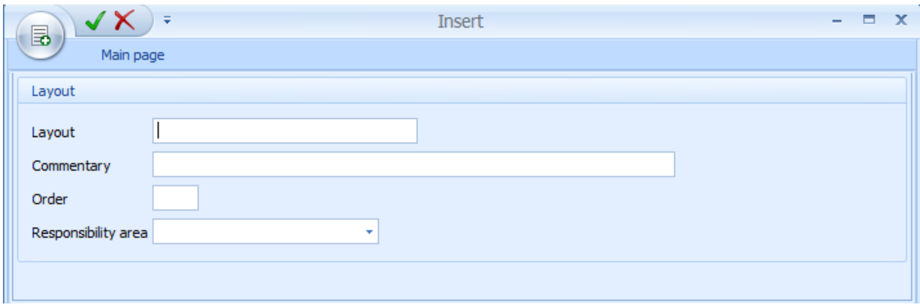
This information refers to the last time a layout entry of the table was edited, and not to the last time the graphic layout was edited.

Toolbar

Insert

Function authorization: mparkle.create

Add a new entry to create a new layout. Please provide the following information: -

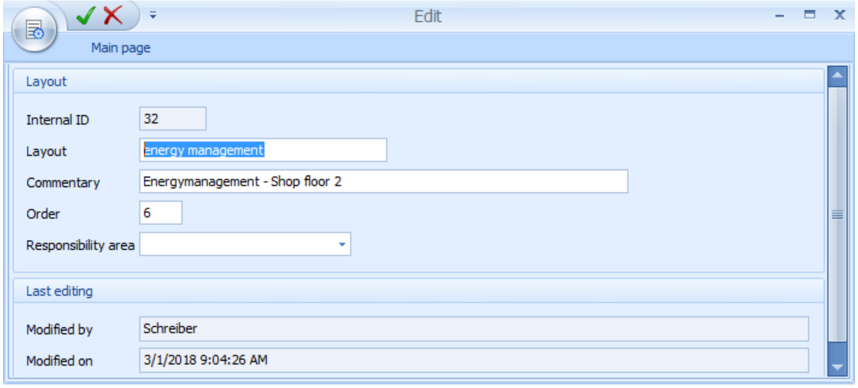


- Layout: Layout name
- Comment: Description
- Order: you can sort the list by this column
- Responsibility area (optional)

Editing

Function authorization: mparkle.edit

You may change the following data in an existing layout



- Layout
- Comment
- Order
- Responsibility area

Delete

Function authorization: mparkle.delete

The system deletes the layout entry from the database, once you have confirmed the deletion.

Please note: The file that saves the actual layout information on the server is not deleted.

Layout editor

Function authorization: mparkle.layedit

Calls up the Layout editor for the entry currently selected in the list.

Full screen

Zooms in the active layout to the full screen size. Press "Esc" to exit the full screen view.

Zoom out

Zooms out the current layout in the application.

Zoom in

Zooms in the current layout in the application.

Search

Use these two drop-down list boxes to search for specific objects in layouts. Select the object type (workplaces/machines, groups, line groups, material buffers, resources, terminals, accesses and access groups) in the upper field. Select the number of the searched object in the lower field.

Once you have selected the object, the list of available layouts indicates all layouts containing the relevant object. The application shows one of these layouts automatically. The searched object is highlighted with a red frame.

When you save a layout, the application identifies the objects included in the layout. You might have to save older layouts once more in order for the application to select and highlight the included objects.

You can only use the function "searching for objects" if you enable the upgrade GraphicMachinery2Search.

The graphic machinery does not support the functions "Print preview" and "Print all".

Selection criteria

Selection criteria are not available in the graphic machinery. When you start the application, the application loads all configured layouts automatically and displays the first layout in the list.

If you specify the layout order and sort the list using the "order" column, the application always shows a particular layout by default when you call up the function. Please note that the defined order is not user-related.

Additional features of the graphic machinery

Opening full screen layouts

Use the menu editor to create an entry opening a defined layout in normal view or full-screen mode in the menu of a specific MOC client (menu depending on the user --> USER scope).

You can only use this function if the menu editor and the corresponding function authorization *mparklink* are available.

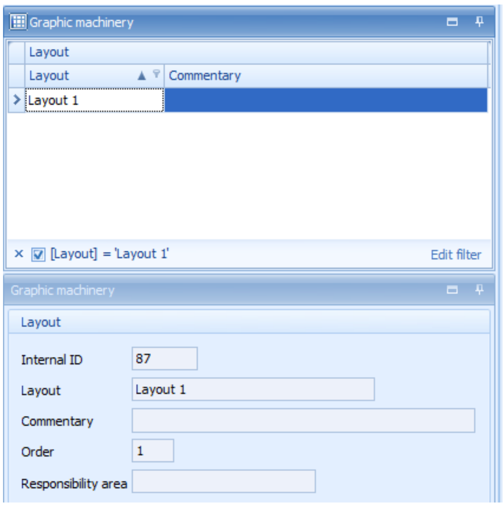
Proceed as follows if the menu editor is used:

1. Open the menu editor:
   Menu --> Extras --> Menu editor
2. Drag the graphic machinery to the required menu and change the values (enable advanced settings).
   - **Caption:** select individually, e.g. Mpark layout 1
   - **ID**: GraphicMachinery2

- **Command**: mpark
- **Parameter**: fullscreen id=<Internal ID of the layout>

  The parameter "fullscreen" opens the defined layout in full screen mode. The details pane of the graphic machinery shows the internal ID of the layout.

Example: “Layout 1” has the internal ID 87 (see screenshot).



Menu editor

If you use the "auto start" function, you can open a defined layout in full screen mode when starting the MOC.

Auto start: right click and select *Add to autostart*.

Setting the refresh interval

The refresh interval refers to the time interval between two update runs in the graphic machinery. You can set the interval via an INI configuration.

The value corresponds to the time interval in milliseconds between two update runs.

The default value of 60000 milliseconds (60 seconds) applies, if you enter a value less than 60000 milliseconds (60 seconds) or if nothing is entered.

Configuration:

INI configuration name:	MPARK
Section:	MPARK2
Key:	REFRESHRATE
Value:	240000 (corresponds to 4 min)

Disabling offline messages

If the graphic machinery cannot establish a connection to the server (offline), the application displays an error message every time you request data.

You might not want this behavior, especially if you use the full screen mode. Configure the INI configuration as follows to prevent the error message from being displayed:

Configuration:

INI configuration name:	MPARK
Section:	MPARK2
Key:	SUPPRESSCONNECTIONERRORMESSAGE
Value:	y
	Default value: n

If you start the graphic machinery the next time and you cannot request data (as a connection to the server cannot be established), the system only records entries in the MOC Log window.

But to inform the user, the window title of the full screen indicates the note "(Offline)" behind the layout name.

Automatic layout change

You can only activate the automatic layout change if you enable the extensions mpark2layrot.

In the graphic machinery, you can change layouts automatically in the full-screen mode. You can configure the display duration and the order of changing the layouts.

<u>Select the layouts and specify the display duration</u>

Hold the CTRL key down to select several layouts at once in the table. Then click the button "configuration" in order to define the display duration.

A dialog opens where you can configure the display duration in seconds for a layout. The least interval for the automatic change is 30 seconds.

Once you have confirmed the dialog, the application stores the layouts and the display duration in a configuration file (LayoutRotation.config) in the MOC user directory.

Starting and stopping the layout change

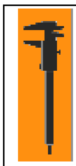
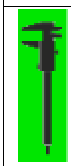
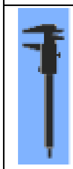
Click the button *Start* to start changing the layouts. After reading the stored settings, the system opens the full screen. Then the application changes layouts in full screen mode at the specified interval.

The time it takes to display the next layout depends on the number of objects displayed in the layout.

Click the button *Stop* or close full screen mode to stop changing the layouts automatically.

Objects "Inspection" and "Inspection points"

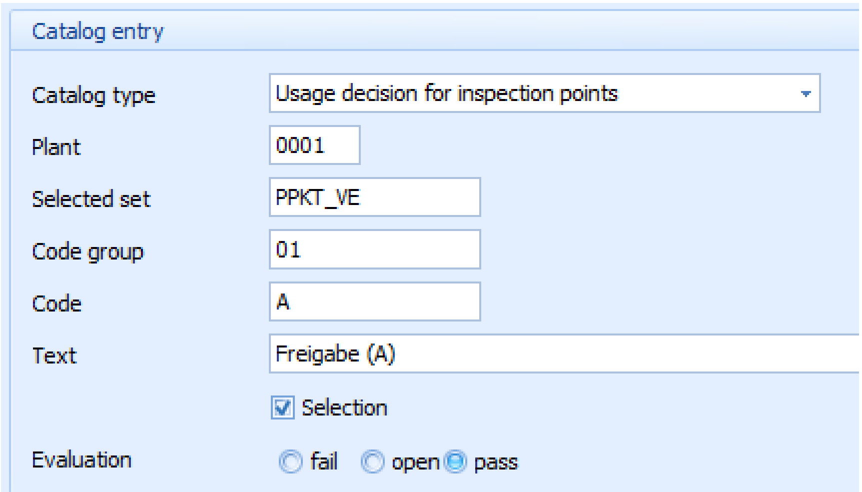
If the in-production inspection is activated and data is collected for inspection points, the current inspection status can also be integrated. To show the inspection status, select the object "inspection" (caliper symbol). The background color illustrates the different inspection statuses.

	Inspection status = due An operation including an inspection step has been logged on to the machine/workplace. At least one inspection point is available for this inspection step.
	Inspection status = checked An operation including an inspection step has been logged on to the machine/workplace. For this inspection step, no inspection point is available.
	Inspection status = inspection not possible The operation logged on to the machine/workplace does not include an operation step or no operation is logged on to the machine/workplace. Machines/workplaces of workplace type "F – in-production inspection" are inspection stations. To this type of machines/workplaces, no operations are logged on. These machines/workplaces always have a blue "inspection status".

	Inspection status = error If the system cannot identify any inspection step for a logged on operation because of an error, then the inspection status is displayed in red.

You can not only show the inspection status, but you can also show the number of open or closed inspection points for the logged on operation. Select the object "inspection points" to this end. With the number of closed inspection points, the system also displays the number of pass and fail inspection points. Condition: with the operation, an inspection step must have been logged on to the machine/workplace. The classification of an inspection point as pass or fail is based on the inspection point decision.

You define the inspection point decisions in the application *Catalog* of the quality management master data using the catalog type *Usage decision for inspection points*. The screenshot below illustrates the possible evaluations of an inspection point.



The screenshot below illustrates the entries for the object "inspection points".